Time Capsule Design Challenge

Unit 10: Time Capsule Engineers · Lesson 10-1 · Flagship · EduWonderLab

UNIT 10 · LESSON 10-1 ★ FLAGSHIP · TEACHER NOTES — **Standard 6.G.2**

FORMAT **Design + compare challenge**	TIME **30–35 min**
GROUPING **Teams of 3–4**	TOPIC **Volume with Whole Number Edges (Flagship)**
STANDARD **6.G.2**	PREP LEVEL **Medium — print design brief, grid paper, dimension cards**

Learning goal *I can design a rectangular-prism capsule, find its volume, and compare designs.*

Teacher setup

- Print one design brief and dimension set per team.
- Teams design a capsule with whole-number dimensions and find its volume.
- They compare which design holds more under the given constraints.

Materials

Design brief, Grid paper, Dimension cards, Volume comparison sheet, Pencil.

Differentiation & language support

- Grid paper helps sketch and count dimensions (SPED).
- Sentence stem: "My capsule holds ___ cubic units because ___."
- ESOL: provide a finished example capsule to model.

Key vocabulary (English / Español):

Term	Español	What it means
dimensions	dimensiones	length, width, and height
capacity	capacidad	how much a container holds

Quick teacher check: *Check the height found on Card 4 (h = 4).*

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Name: **Date:** **Class:**

Your goal: *I can design a rectangular-prism capsule, find its volume, and compare designs.*

What you need

Design brief, Grid paper, Dimension cards.

How to play

1. Read the design task and sketch your capsule on grid paper.
2. Find the volume using length × width × height.
3. Compare your volume with another design.
4. Justify which capsule holds more.

Design Tasks

Design a capsule with whole-number edges, find its volume, and compare it to others.

1. Capsule A is 4 × 3 × 5. Volume?	**2.** Capsule B is 6 × 2 × 5. Volume?
3. Which holds more: 5 × 4 × 3 or 6 × 3 × 4?	**4.** A capsule has volume 48 and a 4 × 3 base. Find the height.
5. Design a capsule with whole edges and volume exactly 36.	**6.** Challenge: Two capsules with volume 48 but different shapes.

Design log (dimensions + volume):

Explain your thinking

Explain how two capsules can have the same volume but different shapes.

Sentence frame: *My capsule holds ___ cubic units because ___ × ___ × ___ = ___.*

★ **Challenge:** *Design the capsule with the greatest volume using edges that add up to 12 (length + width + height = 12).*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Capsule A is 4 × 3 × 5. Volume? **Answer: 60 cubic units**	**2.** Capsule B is 6 × 2 × 5. Volume? **Answer: 60 cubic units (same as A)**
3. Which holds more: 5 × 4 × 3 or 6 × 3 × 4? **Answer: 6 × 3 × 4 = 72 > 60**	**4.** A capsule has volume 48 and a 4 × 3 base. Find the height. **Answer: 4 (48 ÷ 12)**
5. Design a capsule with whole edges and volume exactly 36. **Answer: e.g., 3 × 3 × 4**	**6.** Challenge: Two capsules with volume 48 but different shapes. **Answer: e.g., 4 × 3 × 4 and 8 × 2 × 3**

Teaching notes & look-fors

- Cards 1 and 2 both have volume 60 with different dimensions.
- Card 4: 48 ÷ (4 × 3) = 4.